

3 Sequences, functions and graphs

	Students should be taught to:	Notes
3.1 Sequences	A understand and use common difference (d) and first term (a) in an arithmetic sequence	e.g. given 2nd term is 7 and 5th term is 19, find a and d
	B know and use n th term $= a + (n - 1)d$	
	C find the sum of the first n terms of an arithmetic series (S_n)	e.g. given $4 + 7 + 10 + 13 + \dots$ find sum of first 50 terms
3.2 Function notation	A understand the concept that a function is a mapping between elements of two sets	
	B use function notations of the form $f(x) = \dots$ and $f: x \mapsto \dots$	
	C understand the terms 'domain' and 'range' and which values may need to be excluded from a domain	$f(x) = \frac{1}{x-2}$ exclude $x = 2$
	D understand and find the composite function fg and the inverse function f^{-1}	'fg' will mean 'do g first, then f'
3.3 Graphs	A recognise, plot and draw graphs with equation: $y = Ax^3 + Bx^2 + Cx + D \text{ in which:}$ (i) the constants are integers and some could be zero (ii) the letters x and y can be replaced with any other two letters or: $y = Ax^3 + Bx^2 + Cx + D + \frac{E}{x} + \frac{F}{x^2}$ in which: (i) the constants are numerical and at least three of them are zero (ii) the letters x and y can be replaced with any other two letters or:	$y = x^3$ $y = 3x^3 - 2x^2 + 5x - 4$ $y = 2x^3 - 6x + 2$ $V = 60w(60 - w)$ $y = \frac{1}{x}, x \neq 0,$ $y = 2x^2 + 3x + \frac{1}{x},$ $x \neq 0,$ $y = \frac{1}{x}(3x^2 - 5),$ $x \neq 0,$

	Students should be taught to:	Notes
	B apply to the graph of $y = f(x)$ the transformations $y = f(x) + a$, $y = f(ax)$, $y = f(x + a)$, $y = af(x)$ for linear, quadratic, sine and cosine functions	
	C interpret and analyse transformations of functions and write the functions algebraically	
	D find the gradients of non-linear graphs	By drawing a tangent
	E find the intersection points of two graphs, one linear (y_1) and one non-linear (y_2), and recognise that the solutions correspond to the solutions of $(y_2 - y_1) = 0$	The x values of the intersection of the two graphs: $y = 2x + 1$ $y = x^2 + 3x - 2$ are the solutions of: $x^2 + x - 3 = 0$ Similarly, the x values of the intersection of the two graphs: $y = 5$ $y = x^3 - 3x^2 + 7$ are the solutions of: $x^3 - 3x^2 + 2 = 0$
	F calculate the gradient of a straight line given the coordinates of two points	Find the equation of the straight line through (1, 7) and (2, 9)
	G find the equation of a straight line parallel to a given line; find the equation of a straight line perpendicular to a given line	Find the equation of the line perpendicular to $y = 2x + 5$ through the point (3, 7)
3.4 Calculus	A understand the concept of a variable rate of change	
	B differentiate integer powers of x	
	C determine gradients, rates of change, stationary points, turning points (maxima and minima) by differentiation and relate these to graphs	Find the coordinates of the maximum and minimum points
	D distinguish between maxima and minima by considering the general shape of the graph only	

	Students should be taught to:	Notes
	E apply calculus to linear kinematics and to other simple practical problems	The displacement, s metres, of a particle from a fixed point O after t seconds is given by: $s = 24t^2 - t^3,$ $0 \leq t \leq 20$ Find expressions for the velocity and the acceleration.